Adam O'Deven

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Mechanical Engineering and Industrial Design student at Georgia Tech with extensive hands-on experience. Proven skills in owning projects from concept through validation, extracting insights through prototyping, and working across the design–engineering handoff in cross-functional teams.

EDUCATION

Georgia Institute of Technology – BS Mechanical Engineering, BS Industrial Design · 3.6 GPA · Atlanta · Jun 2024 – Present
National University of Singapore – Industrial Design exchange semester · Singapore · Fall 2026
Georgia Tech Europe – Mechanical Engineering study abroad · Metz, France · Summer 2025

SKILLS

Mechanical Design – SolidWorks, Fusion, parametric modeling, top-down design, packaging, DFM/DFA, GD&T familiarity
Prototyping – CNC machining (3-axis), SLS/FDM/SLA printing, manual mill, metal lathe, assembly, soldering, circuit design
Design & Visualization – Concept sketching, KeyShot, Vizcom, Adobe Suite, product photography, appearance models
Research & Validation – Field observation, user interviews and surveys, usability testing, prototype testing, design iteration

EXPERIENCE

Scout Motors · Novi, MI · May 2026 – Aug 2026 / ***Advanced Concepts Engineering Intern***

- Owned an end-to-end pedestrian protection concept program, advancing five concepts through research, decision matrix down-selection, and two functional prototypes to final review with engineering leadership, yielding three invention disclosures and one patent filing.
- Defined the concept space from three competing constraints (NCAP lower-leg injury metrics, best-in-class approach angle, heritage inspired front-end design intent) and generated five concepts satisfying all three.
- Consulted CAE, Vehicle Architecture, V&V, Safety, and Legal throughout concept development, translating their domain requirements into design decisions across all five concepts.
- Rebuilt the prototype lab's part-request workflow with missing-information and print-approval gates, then produced 293 in-house parts for a prototype vehicle build, avoiding $75,000+ against outsourced quotes.
- Provided DFM support and presented parts to the production team in Columbia, SC, securing per-part quality sign-off.

REAR Lab · Atlanta, GA · Aug 2024 – Dec 2024 / ***Product Developer***

- Derived mounting, reach, and clearance requirements for a chair-mounted dog treat dispenser by fitting prototypes to power wheelchairs in use at the Shepherd Center.
- Rotated the lid direction and lowered the dispenser after on-chair testing, clearing the user's natural hand position.
- Fabricated each revision from 3D-printed, manually machined, and off-the-shelf parts, selecting efficient processes part-by-part to enable three full user feedback rounds in one semester.
- Revised the dispenser across 5+ builds and three rounds of Shepherd Center user feedback, relocating the mount into seated reach, reducing actuation force, and refining the dispensing path to reduce jamming across treat sizes.
- Evaluated and documented SolidWorks against Fusion for made-to-order products, giving the lab a basis for tool selection on later projects.

RoboJackets K12 Foundation · Atlanta, GA · Jun 2018 – Aug 2026 / ***Mentor, Mechanical Engineer, and Brand Designer***

- Led mechanical design, prototyping, and assembly of 28 subsystems across seven competition robots in four seasons, contributing to 13 event wins and world championship qualification every season.
- Owned the complete mechanical design of a second competition robot, developing a custom high-speed drivetrain and vacuum downforce system that took the robot to a division win.
- Built parametric CAD assemblies for both prototype and competition systems, propagating subsystem changes through the robot without downstream rework.
- Integrated mechanical and electrical subsystems within tight spatial constraints, maintaining modularity for quick repairs.

PROJECTS

SpeedCox ↗ · Class Project · Spring 2026 / ***race instrument for rowing coxswains***

- Derived the feature set from a varsity coxswain interview, nine rower surveys, and a teardown of competing rowing electronics, mapping pain points across seven race stages into prioritized design opportunities.
- Added a real-versus-goal pace meter beneath the primary read so the coxswain judges pace as a position rather than comparing a raw split against a goal held in memory mid-race.

Beacon ↗ · Class Project · Fall 2025 / ***emergency/disaster relief medical light***

- Sized the pressure vessel from payload mass upward, calculating helium volume against net lift to land a 0.2 L canister small enough to carry on a belt.
- Cut light payload from 50 g to 27 g by stripping an off-the-shelf fixture to its bare emitter and relocating the cell to the handle, halving required gas storage from 0.24 L to 0.13 L.
- Packaged tank, balloon, light module, and tether into a 1.94″ diameter by 5.87″ tall body deployed by a single pull tab.

Card Failure Design Challenge ↗ · Application Material · Dec 2025 / ***problem analysis and machine design***

- Observed grip variation across 10 users to set force application points, which established the requirement for a variable-position load head.
- Bounded force-to-yield at 63.4 N under nominal material properties and 37.85 N at worst-case published values, identifying single overload rather than fatigue as the dominant failure mode.